

### Project-based assessment

Student master the fundamentals of React, be a proficient React developer, be able to create apps independently, and make use of the appropriate React.

Student demonstrate their skills in designing user interfaces, dealing with APIs, and maintaining the state of a React app. got practice using Postman, became really comfortable using React.

Each student develops a project using React.js under Node.js. Several key expectations should be considered. The following are minimum some highlights:

- Knowledge of Node.js fundamentals, including setting up a server using Express.js.
- Ability to handle routing, middleware, and API endpoints.
- Correct use of React concepts (components, state, props, hooks)
- Knowledge of core React concepts such as components, state, props, and lifecycle methods.
- Ability to create functional components and class components.
- Understanding of hooks (e.g., `useState`, `useEffect`).
- Effective state management (using Context API.)
- Proper routing implementation (React Router usage)
- Routes and controllers using ExpressJs, NodeJs
- Use of Responses and Request Objects in Express JS.
- Skills in making HTTP requests using `fetch` or Axios to communicate with a Node.js backend.
- Understanding RESTful principles and how to consume APIs.
- API integration and data handling (fetching and displaying data) API development calls with Fetch or Axios and JSON parsing to obtain data.
- Create an API which could handle CRUD actions.
- Format and make HTTP requests and view the response.
- Dynamically populate the data list into a HTML table, List, Paragraphs, divisions with user queries
- Building at least 4 web pages application in React (Form, List, Table, Divisions)

By focusing on these areas, students can effectively develop robust applications using React.js and Node.js, gaining valuable skills in modern web development.

Due Date: Submission of a project must be uploaded on the Moodle before this date 07-11-2024.

Please note that your presentation be face to face on campus and **the evaluation will not only be based on the quality of your project but also on how well you respond to questions during the presentation.**

Additionally, please remember that you are responsible for all materials covered in class sessions, including online sessions until 07-11-2024. Make sure you review all relevant content to prepare thoroughly.

Start simply, create core functionality, and then add features gradually.